

MATTHEW HEFFERNAN, PHD

✉ mrhheffernan@gmail.com

☎ 628-255-0602

in [linkedin.com/in/mrhheffernan](https://www.linkedin.com/in/mrhheffernan)

🐙 github.com/mrhheffernan

Scientist specializing in quantitative modeling, large-scale scientific computing, and AI verification with expertise building reliable, high-performance statistical tools.

SKILLS

Probability & Statistics Bayesian inference | Uncertainty quantification | Experimental design | Sensitivity analysis
Research Design of Experiments | Statistical modeling | Data Science | Data visualization | Scientific computing
Programming *Proficient:* Python | Bash | Linux/Unix | *Developing:* SQL | Rust | *Other:* Code review | Software testing
Tools Databricks | PySpark | Bazel | Scikit-learn | MCMC | Git | Jupyter | Airflow | Pandas | SciPy | matplotlib | PyO3

CAREER AND PROFESSIONAL EXPERIENCE

AI Verification Engineer

Zoox

09/2022 – Present

- Conducted applied research on a methods team, developing and deploying novel black-box safety quantification techniques for the AI driving stack.
- Built tools, automated processes, and data products using Python, Databricks, and PySpark to support autonomous driving evaluation, contributing to 9 patent applications (2 granted)
- Designed a data-collection strategy, developed fleet simulations, and built a test asset orchestration system that enabled scaling and release of the Zoox service to the public.
- Drove continuous improvements to orchestration system resulting in over 99% uptime, scale testing customer-facing ridehail systems and enabling data collection for major company releases.
- Decomposed complex problems into known statistical problems, developed prototypes, and deployed tooling to orchestrate the test vehicle fleet and deliver \$25M+ of log data for safety validation and AI development.

Developed production statistical analysis research

as Research Assistant at McGill University

09/2016 – 08/2022

- Trained, validated, and deployed models for dimensionality reduction, regression, and statistical inference on several hundred gigabytes of high-dimensional data
- Improved computational study design performance by over 50% for model prediction, sensitivity analysis, and uncertainty quantification using advanced statistical algorithms for design spaces robust to inactive dimensions
- Developed, validated, and automated C++ and Python scientific-computing tools for data analysis, streamlining the workflow and reducing manual job orchestration
- Contributed to developing and validating statistical analysis for the most advanced C++ framework of a sophisticated nuclear physics environment (jetscape.org)
- Communicated results and analytics effectively via expert/non-expert presentations, publications, documentation, and workshops to stakeholders and technical experts
- Published a proof-of-concept study for physics laboratory experiment design using Bayesian parameter estimation and model selection, delivering actionable pedagogical guidance for laboratory instructors.
- Managed research group computing resources, supported collaborative projects, and mentored new researchers

EDUCATION

Ph.D. Physics, "Quantification of the Quark-Gluon Plasma with Statistical Learning"

McGill University

08/2018 – 10/2022

M.Sc. Physics

McGill University

09/2016 – 08/2018

B.Sc. Physics (Hon.), German Studies minor

The College of William & Mary

08/2012 – 05/2016